

LOW ACRYLONITRILE NBR: PATENT LANDSCAPE ON COMPOSITION CONTROL, EMULSION POLYMERIZATION, AND POST-POLYMER MODIFICATION

1. ABSTRACT

This report synthesizes six primary patents relevant to low-acrylonitrile nitrile butadiene rubber (NBR), focusing on how acrylonitrile content, monomer feed strategy, and downstream processing influence rubber performance. The target material is low-ACN NBR, generally understood as NBR with reduced acrylonitrile content relative to conventional grades, often in the approximate 10–18 wt% range, where composition control is critical for balancing oil resistance, low-temperature flexibility, and processability. The patent set spans two major synthesis themes: direct copolymer preparation by emulsion or solution polymerization, and post-polymer modification/degradation routes that alter molecular weight or functionality without changing the base NBR backbone.

US9650502B2 is the most directly relevant to low-ACN NBR preparation, disclosing emulsion polymerization of acrylonitrile and butadiene with staged acrylonitrile feeding, alkyl thiol chain-transfer control, and pH-managed latex coagulation/washing. The examples explicitly show low initial acrylonitrile charges followed by continuous feed, conversion-controlled booster additions, and coagulation with magnesium sulfate followed by alkaline washing. EP2145681B1 and US7875683B2 address metathesis-based molecular-weight reduction of NBR, enabling lower-Mn feedstocks and narrower distributions for subsequent hydrogenation or functionalization; these are not composition-setting patents, but they are important for processing low-viscosity NBR derived from controlled copolymer feeds. EP1825913B1 further develops metathesis catalyst systems by adding salts such as LiBr or CsBr to accelerate ring-closing metathesis, with broader implications for NBR metathesis efficiency. US10011664B2 discloses metathesis functionalization of nitrile rubbers using functionalized allyl compounds and ruthenium/osmium carbene catalysts, showing a route to modified NBR materials. EP2727948B1 is less directly related to acrylonitrile content, but it demonstrates solution polymerization with neodymium catalyst systems and controlled monomer feed in conjugated diene copolymerization, reinforcing the broader process-control paradigm used in elastomer design.

Overall, the evidence favors semi-continuous emulsion polymerization with controlled acrylonitrile feed, thiol chain transfer, and carefully managed coagulation/washing as the most practical route

to low-ACN NBR, while metathesis patents provide complementary routes for molecular-weight tailoring and downstream functionalization.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: US9650502B2
- Patent Title: Process for the preparation of nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses emulsion polymerization of at least one α,β -unsaturated nitrile and at least one conjugated diene, optionally with additional copolymerizable monomers, in the presence of a C12–C16 alkyl thiol chain-transfer agent having at least three tertiary carbon atoms. The resulting latex is coagulated with a metal sulfate and then washed with water containing an inorganic base at elevated pH to obtain nitrile rubber with improved vulcanization behavior and reduced mold fouling. - pH of emulsion polymerization: 7 to 10 — Abstract/Claim 1 - Preferred pH of emulsion polymerization: 8 to 9 — Claim 2 - Alkyl thiol carbon range: 12 to 16 carbon atoms — Abstract/Claim 1 - Alkyl thiol tertiary carbon atoms: at least 3 — Abstract/Claim 1 - Coagulation temperature: ≥ 30 °C — Abstract/Claim 1 - Preferred coagulation temperature: 40 to 60 °C — Claim 4 - Washing pH: ≥ 10 — Abstract/Claim 1 - Preferred washing pH: 11 to 12 — Abstract/Claim 1 - Autoclave volume: 70 liter — Example 1 - Deionized water charge: 9.92 kg — Preparation of Latex A - Potassium stearate solution charge: 0.53 kg of 5.4 wt% solution — Preparation of Latex A - DNMK charge: 0.440 kg — Preparation of Latex A - pH adjusted to: 8.5 — Preparation of Latex A - Nitrogen flush: used — Preparation of Latex A - Acrylonitrile initial charge: 500 g — Preparation of Latex A - TDM charge: 18.72 g — Preparation of Latex A - Butadiene charge: 4.37 kg — Preparation of Latex A - Polymerization temperature: 7 °C — Preparation of Latex A - DIHP charge: 6 g — Preparation of Latex A - FSS solution charge: 139 g — Preparation of Latex A - Remaining acrylonitrile feed: 1.134 kg — Preparation of Latex A - Acrylonitrile feed rate: 0.162 kg/h — Preparation of Latex A - First booster TDM addition: 2.34 g at 37% conversion — Preparation of Latex A - Second booster TDM addition: 2.34 g at 59% conversion — Preparation of Latex A - Polymerization stop at: 70% conversion — Preparation of Latex A - Polymerization time: 7 h — Preparation of Latex A - Hydroxylamine sulfate solution added: 515 g of 2 wt% solution — Preparation of Latex A - Coagulation salt: saturated aqueous calcium chloride — Examples 7-9 comparative - Washing pH

after CaCl₂ coagulation: 11.5 to 11.6 — Examples 7-9 comparative

Relevant Experimental Evidence - Example 1: Latex A was prepared by emulsion polymerization at 7 °C with staged acrylonitrile feed and thiol booster additions; the resulting NBR was later evaluated for vulcanization and mold fouling. - Example 2: Latex B was prepared under similar conditions with a different monomer/thiol balance, enabling comparison of composition and processing behavior. - Example 3: Latex A/B/C formulations and latex preparation details were disclosed, including monomer charges, emulsifier, chain-transfer agent, initiator, reducing agent, and shortstop system. - Example 4: Comparative NBR obtained after neutral-pH washing showed poorer vulcanization rate, lower yield, and mold fouling. - Example 6: NBR from Latex C also showed inferior rheometric behavior and mold fouling relative to the invention process. - Example 7: Calcium chloride coagulation followed by alkaline washing produced NBR with worse vulcanization performance and severe mold fouling in comparative testing.

Technical Relevance This is the most directly relevant patent for low-acrylonitrile NBR because it discloses controlled emulsion copolymerization of acrylonitrile and butadiene with staged monomer feed and explicit composition/process control. The examples show how latex preparation, coagulation, and washing conditions affect final rubber performance, which is central to producing low-ACN grades with good processability.

Patent 2

- Patent Number: US10011664B2
- Patent Title: Functionalized nitrile rubbers and the production thereof
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2018
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses metathesis-based functionalization of nitrile rubbers by contacting NBR with a complex catalyst based on a transition-group 6 or 8 metal carbene and a functionalized allyl compound. The process can introduce terminal functional groups and is described as suitable for producing vulcanizable mixtures and block copolymers.

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance Although it does not define acrylonitrile content, this patent is relevant because it uses nitrile rubber as the substrate and demonstrates a post-polymer modification route that can be applied to NBR derived from low-ACN feedstocks. It is useful for understanding downstream functionalization options after composition-controlled NBR synthesis.

Patent 3

- Patent Number: US7875683B2
- Patent Title: Process for the metathetic degradation of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2011
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses molecular-weight reduction of nitrile rubber by olefin metathesis using a catalyst system based on a transition-group 6 or 8 metal carbene complex together with a transition-metal additive of general formula $M(OZ)_m$. The process is intended to produce lower-Mn NBR suitable for subsequent hydrogenation to low-Mooney HNBR.

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant to low-acrylonitrile NBR because it addresses the processing and molecular-weight tailoring of NBR feedstocks, which is often necessary when low-ACN grades are prepared for hydrogenation or further modification. It provides a route to lower-viscosity NBR without changing the base copolymer composition.

Patent 4

- Patent Number: EP2727948B1
- Patent Title: High cis diene/phenylbutadiene copolymers prepared using a ziegler/natta ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses solution polymerization of 1,3-butadiene with substituted butadienes using a neodymium-based Ziegler-Natta catalyst system formed from an organoaluminum compound, a neodymium carboxylate, and a dialkyl aluminum chloride. It emphasizes stereoregular copolymer formation and controlled monomer feed in hydrocarbon solvent. - 1-Phenylbutadiene in monomer feed: 1.2 wt% relative to butadiene — Example 2 - Polymerization temperature: 65 °C — Example 2 - Polymerization time: 1 h — Example 2 - Catalyst solution: 0.42 mL of 0.0191 M 1/25/2.5 Nd Versatate/TIBA/DEAC — Example 2 - TIBA solution: 0.8 mL of 0.1 M in hexanes — Example 2 -

Butadiene solution concentration: 15% in hexanes — Example 2 - Monomer feed in polymerization medium: 75 to 99.5 wt% butadiene — Detailed Description - Monomer feed in polymerization medium: 0.5 to 25 wt% phenylbutadiene — Detailed Description - Neodymium catalyst level: 0.05 to 1.0 millimole Nd per 100 g monomer — Detailed Description - Preferred neodymium catalyst level: 0.1 to 0.3 millimole Nd per 100 g monomer — Detailed Description - Catalyst preformation temperature: 0 to 100 °C — Detailed Description - Preferred catalyst preformation temperature: 10 to 60 °C — Detailed Description - Preferred catalyst preformation temperature: 15 to 30 °C — Detailed Description - Neodymium-aluminum catalyst component reaction time: 12 to 30 min — Detailed Description - Preferred neodymium-aluminum catalyst component reaction time: 15 to 25 min — Detailed Description - Dialkyl aluminum chloride reaction time: at least 30 min — Detailed Description - Catalyst activity improvement time: up to about 24 h — Detailed Description

Relevant Experimental Evidence - Example 1: 1-phenylbutadiene was synthesized from trans-cinnamaldehyde using n-butyl lithium in THF at 0 °C, then isolated by distillation at 62 °C and 6 mbar. - Example 2: Butadiene/1-phenylbutadiene copolymerization at 65 °C for 1 h gave 90% yield and Mn 162,000 with 1.2 wt% comonomer. - Example 2: Increasing phenylbutadiene content reduced reaction rate and altered molecular weight, showing feed-composition sensitivity. - Example 2: The copolymer retained high cis-1,4 microstructure (98.0%), demonstrating stereoregular solution polymerization. - Example 2: The catalyst system was effective at low neodymium loading and in hydrocarbon solvent.

Technical Relevance This patent is only indirectly relevant to low-acrylonitrile NBR because it does not use acrylonitrile, but it is valuable as a model for controlled solution copolymerization and catalyst design in diene elastomers. The feed-control and catalyst-preformation concepts are transferable to composition-controlled NBR synthesis.

Patent 5

- Patent Number: EP2145681B1
- Patent Title: Method for the reduction of the molecular weight of nitrile rubber
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2013
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses molecular-weight reduction of nitrile rubber by metathesis using a catalyst system that combines a metathesis catalyst with a compound of general formula $\text{BF}_m\text{X}_n\text{D}_v$, where BF_3 -derived species are paired with donor molecules such as water, ether, amines, alcohols, or acids. The process is intended to improve degradation efficiency and produce lower-molecular-weight NBR.

Relevant Experimental Evidence - No experimental evidence extracted from available evidence.

Technical Relevance This patent is relevant to low-acrylonitrile NBR because it addresses the downstream molecular-weight tailoring of NBR, which is important when low-ACN grades are used as feedstocks for hydrogenation or functionalization. It complements composition-control patents by improving processability through degradation chemistry.

Patent 6

- Patent Number: EP1825913B1
- Patent Title: New catalyst systems and their application for metathesis reactions
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: The patent discloses metathesis of nitrile rubber using a catalyst system comprising a metathesis catalyst and one or more salts, with examples showing salt-accelerated ring-closing metathesis. The examples demonstrate that LiBr and CsBr can accelerate metathesis conversions under mild conditions. - LiBr addition: 1 mg — Example 7.02 - CsBr addition: 1 mg — Example 8.02 - Grubbs II catalyst charge: 10 mg — Example 7 - Chlorobenzene charge: 0.3 mL — Example 7 - CDCl₃ charge: 0.2 mL — Example 7 - Diethyl diallylmalonate addition: 0.15 mL every 2 min — Example 7 - Reaction temperature: room temperature — Example 7 - LiBr accelerated conversion to: 55.4% at 30 min and 100% at 60 min — Example 7.02 - Without salt conversion: 21.3% at 30 min and 57.7% at 60 min — Example 7.01 - CsBr accelerated conversion to: 16.5% at 15 min, 40.3% at 30 min, 68.9% at 60 min, 84.7% at 90 min, 100% at 150 min — Example 8.02 - Without salt conversion: 13.4% at 15 min, 25.3% at 30 min, 46.5% at 60 min, 71.9% at 90 min, 96.2% at 150 min — Example 8.01

Relevant Experimental Evidence - Example 7: LiBr accelerated ring-closing metathesis of diethyl diallylmalonate relative to the no-salt control. - Example 8: CsBr also accelerated ring-closing metathesis, though less strongly than LiBr in the reported time window.

Technical Relevance This patent is relevant to low-acrylonitrile NBR because it improves metathesis catalyst performance on nitrile rubber substrates, which is useful for molecular-weight reduction and related post-polymer modifications. It provides a practical catalyst-additive strategy for processing NBR-derived materials.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- US9650502B2 is the only patent in the set that directly controls NBR composition during polymerization; it uses emulsion polymerization with staged acrylonitrile feed, whereas the other patents focus on post-polymer metathesis or unrelated diene copolymerization.
- US9650502B2 emphasizes low-temperature emulsion polymerization at 7 °C with continuous acrylonitrile feed and thiol chain transfer, while EP2727948B1 uses solution polymerization at 65 °C with a neodymium catalyst system and no acrylonitrile.
- The metathesis patents (US7875683B2, EP2145681B1, EP1825913B1, US10011664B2) are process-modification patents rather than composition-setting patents; they target molecular-weight reduction, catalyst efficiency, or functionalization of existing NBR.
- US7875683B2 and EP2145681B1 both address NBR degradation by metathesis, but US7875683B2 adds transition-metal compounds of formula $M(OZ)_m$, while EP2145681B1 uses BF₃-derived additive complexes with donor molecules.
- EP1825913B1 focuses on salt-promoted metathesis and explicitly shows LiBr and CsBr accelerating conversion, whereas US7875683B2 and EP2145681B1 focus on catalyst-system composition rather than salt acceleration.
- US10011664B2 differs from the degradation patents by using metathesis for functionalization with allyl compounds, indicating a route to modified NBR rather than lower-Mn NBR.
- Across the set, the most explicit composition-control variables are monomer feed ratio, staged addition, conversion control, and pH management in US9650502B2; the other patents mainly disclose catalyst systems, temperatures, and reaction times for downstream transformations.

4. CONCLUSION

The evidence indicates that low-acrylonitrile NBR is best approached through controlled emulsion copolymerization of acrylonitrile and butadiene, with staged acrylonitrile feeding, low-temperature operation, and chain-transfer control as shown most clearly in US9650502B2. For downstream processing, metathesis-based patents provide useful tools for lowering molecular weight, narrowing distribution, or introducing functionality, but they do not replace composition control during polymerization. Based on the disclosed data, the most defensible synthesis strategy is semi-continuous emulsion polymerization with explicit acrylonitrile feed control, pH around 8–9 during polymerization, and coagulation/washing under alkaline conditions to obtain processable low-ACN NBR with good vulcanization behavior.

5. REFERENCES

- US9650502B2 | Process for the preparation of nitrile rubbers | Not disclosed | US | 2017 | <https://patents.google.com/patent/US9650502B2/en>
- US10011664B2 | Functionalized nitrile rubbers and the production thereof | Not disclosed | US | 2018 | <https://patents.google.com/patent/US10011664B2/en>
- US7875683B2 | Process for the metathetic degradation of nitrile rubber | Not disclosed | US | 2011 | <https://patents.google.com/patent/US7875683B2/en>

- EP2727948B1 | High cis diene/phenylbutadiene copolymers prepared using a ziegler/natta ... | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2727948B1/en>
- EP2145681B1 | Method for the reduction of the molecular weight of nitrile rubber | Not disclosed | EP | 2013 | <https://patents.google.com/patent/EP2145681B1/en>
- EP1825913B1 | New catalyst systems and their application for metathesis reactions | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP1825913B1/en>